

MICROCALCULATORS "ELEKTRONIKA MK 66" "ELEKTRONIKA MK 66A"

OPERATING MANUAL PAG3.050.006 RE

DEAR COMRADE!

You are now the owner of a domestically made microcalculator.

A microcalculator is a miniature electronic computing device for personal use.

Careful handling of the microcalculator, tidiness and compliance with the requirements of this operating manual will ensure its reliable operation for a long time.

We are confident that the microcalculator will be a reliable helper in your mathematical calculations.

The manufacturer's engineers are continually working to improve the technical and operating characteristics of the microcalculator and will be grateful for your feedback on its performance.

Please send the completed feedback slip to:
432030, Ulyanovsk, "Iskra" Plant, 2 Repina St.

A T T E N T I O N !

When buying the microcalculator (MC), ask for its operation to be checked by demonstrating various operations from the control test given in this manual, both on battery power and on mains power.

Check that the set is complete and that the seals on the microcalculator and the power supply are intact, and that the warranty card and tear-off coupons are present in the operating manual. Make sure they bear the seller's stamp or legible signature, the shop stamp and the date of sale.

Remember: if you lose the warranty card, you lose the right to warranty repair.

Tear-off coupons are cut out by the repair organisation's staff only after the work has actually been done.

Do not begin using the microcalculator until you have read this operating manual carefully.

Do not wipe the front panel or the display of the microcalculator with organic solvents or alcohol-based mixtures.

If the calculator will not be used for a long time, remove the batteries.

USE ONLY A316 "KVANT" OR "PRIMA" BATTERIES!

1. CONTENTS OF DELIVERY

1. "Elektronika MK 66" or "Elektronika MK 66A" microcalculator (without batteries), pcs	1
2. "Elektronika D2-10M" power supply, pcs	1
3. Case, pcs	1
4. Operating manual, copies	1
5. A316 "Kvant" or "Prima" batteries, pcs	3
6. Individual retail packaging, pcs*	1

* Numbered "3" in the original, a misprint.

2. MAIN TECHNICAL SPECIFICATIONS

Operations performed: the four arithmetic operations, two-level bracketed calculations, natural and common logarithms and antilogarithms, square roots, raising any positive real number to a power, reciprocals, factorials, conversion of degrees to radians and back, data storage and memory operations. On the "Elektronika MK 66", whether trigonometric functions take their argument in degrees or radians is set by the $\boxed{\Gamma/P}$ (Degrees/Radians) switch. On the "Elektronika MK 66A", trigonometric functions take their argument in degrees. To evaluate a trigonometric function of an angle in radians, first convert the angle from radians to degrees with the $\boxed{F}$ $\boxed{P \rightarrow \Gamma}$ keys.

On the "Elektronika MK 66", whether inverse trigonometric functions return results in degrees or radians is set by the $\boxed{\Gamma/P}$ switch. On the "Elektronika MK 66A" the result is in degrees. To get the result in radians, convert it from degrees to radians with the $\boxed{F}$ $\boxed{\Gamma \rightarrow P}$ keys.

Arithmetic operation time, s	≤ 0.6
Number format	a) fixed decimal point b) exponential (scientific)
Range of numbers: fixed point	$\pm 10^{-7}$ to $\pm(10^8 - 1)$
exponential	$\pm 10^{-99}$ to $\pm 9.9999999 \cdot 10^{99}$
Power consumption, W	≤ 0.45
Power supply	three A316 “Kvant” or “Prima” cells, or 220 V (+10/–15 %), (50 ± 1) Hz mains via the power supply unit
Operating temperature range, °C	+10 to +35
Overall dimensions, mm	155 × 78 × 28
Weight (without batteries), kg	0.15
Precious metal content: gold, g	0.00326
silver, g	–

PERMITTED INPUT RANGES AND CALCULATION ACCURACY

The microcalculator performs the four arithmetic operations (+, –, ×, ÷), operations with π , square roots and reciprocals with a maximum error of ± 2 in the first digit of the mantissa. For square roots the argument must be in the range $0 \leq x \leq 9.9999999 \cdot 10^{99}$.

WORKED EXAMPLES

$$6.3 + 5.65 = 11.95$$

$$6 \boxed{.} 3 \boxed{+} 5 \boxed{.} 65 \boxed{=}$$

$$11.95$$

$$75 \cdot 10^3 - 8 \cdot 10^{-3} = 74999.992$$

$$75 \boxed{B\pi} 3 \boxed{-} 8 \boxed{B\pi} 3 \boxed{/} \boxed{=}$$

$$74999.992$$

TRIGONOMETRIC FUNCTIONS

The trigonometric functions $\sin x$, $\cos x$ and $\operatorname{tg} x$ ($\tan x$) can be calculated for an argument x in degrees or radians.

All trigonometric results have the correct algebraic sign.

If the argument is in the range $10^{-8} \leq |x| \leq 2\pi$ (360°), or $x = 0$, results have a maximum error of ± 5 in the third digit of the mantissa. Accuracy falls for angles outside this range. The permitted range of x is +25 to -25 periods for sin and tg, and +24¾ to -25¼ periods for cos.

WORKED EXAMPLES

For the “Elektronika MK 66”, with the $\boxed{P/\Gamma}$ switch set to $\boxed{\Gamma}$ (degrees):

$$\sin 30^\circ = 0.5$$

$$30 \boxed{F} \boxed{\sin} \quad 5. \quad -1$$

$$\cos 45^\circ = 0.707107$$

$$45 \boxed{F} \boxed{\cos} \quad 7.07107 \quad -1$$

With the $\boxed{P/\Gamma}$ switch set to $\boxed{P}$ (radians):

$$\text{tg } 1.1 \text{ rad} = 1.96476$$

$$1 \boxed{.} 1 \boxed{F} \boxed{\text{tg}} \quad 1.96476^*$$

For the “Elektronika MK 66A”:

$$\sin 30^\circ = 0.5$$

$$30 \boxed{F} \boxed{\sin} \quad 5. \quad -1$$

$$\cos 45^\circ = 0.707107$$

$$45 \boxed{F} \boxed{\cos} \quad 7.07107 \quad -1$$

$$\sin 0.52359878 \text{ rad} = 0.5$$

$$\boxed{.} 52359878 \boxed{F} \boxed{P \rightarrow \Gamma} \boxed{F} \boxed{\sin} \quad 5. \quad -1$$

$$\cos 0.7854 \text{ rad} = 0.707106$$

$$\boxed{.} 7854 \boxed{F} \boxed{P \rightarrow \Gamma} \boxed{F} \boxed{\cos} \quad 7.07106 \quad -1$$

* The original prints the display as 1.95476, a misprint; $\tan 1.1 = 1.96476$.

INVERSE TRIGONOMETRIC FUNCTIONS

The results of arc sin x , arc cos x and arc tg x are given in degrees or radians, with the correct algebraic sign, in the ranges:

$$-90^\circ (-\pi/2 \text{ rad}) \leq \arcsin x \leq +90^\circ (\pi/2 \text{ rad}),$$

$$0^\circ (0 \text{ rad}) \leq \arccos x \leq +180^\circ (\pi \text{ rad}),$$

$$-90^\circ (-\pi/2 \text{ rad}) \leq \operatorname{arctg} x \leq +90^\circ (\pi/2 \text{ rad}),$$

with a maximum error of ± 5 in the third digit of the mantissa. Argument range for $\arcsin x$ and $\arccos x$: $10^{-9} \leq |x| \leq 1$ or $x = 0$; for $\operatorname{arctg} x$: $10^{-9} \leq |x| \leq 10^9$ or $x = 0$.

WORKED EXAMPLES

For the “Elektronika MK 66”, with the $\boxed{P/\Gamma}$ switch set to $\boxed{\Gamma}$:

$$\arcsin 0.389 = 22.8923^\circ$$

$$\boxed{.} \boxed{389} \boxed{\text{arc}} \boxed{\sin} \quad 22.8923$$

$$\operatorname{arctg} 1 = 45^\circ$$

$$1 \boxed{\text{arc}} \boxed{\text{tg}} \quad 45.$$

For the “Elektronika MK 66A”:

$$\arcsin 0.389 = 0.39954601 \text{ rad}$$

$$\boxed{.} \boxed{389} \boxed{F} \boxed{\text{arc}} \boxed{\sin} \boxed{F} \boxed{\Gamma \rightarrow P} \quad 3.9954601 \quad -1$$

$$\operatorname{arctg} 1 = 0.78539817 \text{ rad}$$

$$1 \boxed{F} \boxed{\text{arc}} \boxed{\text{tg}} \boxed{F} \boxed{\Gamma \rightarrow P} \quad 7.8539817 \quad -1$$

LOGARITHMIC FUNCTIONS

Natural and common logarithms can be calculated for positive numbers in the range $10^{-99} \leq x \leq 9.9999999 \cdot 10^{99}$, with a maximum error of ± 5 in the third digit of the mantissa.

WORKED EXAMPLES

$$\ln 32^3 = 10.39722$$

$$3 \boxed{\times} \boxed{32} \boxed{F} \boxed{\ln} \boxed{=} \quad 10.39722$$

$$\lg 120 = 2.07918$$

$$120 \boxed{F} \boxed{\lg} \quad 2.07918$$

EXPONENTIAL FUNCTIONS

For e^x the argument must be in the range $\ln 10^{-99} < x < 10^{100}$ or $x = 0$; for 10^x , in the range $-99 \leq x \leq 100$. Results are accurate to ± 5 in the third digit of the mantissa.

For y^x the argument must be in the range

$$\ln(10^{-99}) / \ln y \leq x \leq \ln(9.9999999 \cdot 10^{99}) / \ln y,$$

and y must be in the range

$$10^{-99} \leq y \leq 9.9999999 \cdot 10^{99}.$$

The calculation is done in two stages according to the formula:

$$y^x = e^{x \ln y} = 10^{x \lg y}$$

WORKED EXAMPLES

$$e^{-3} = 0.0497871$$

$$3 \left[\frac{-}{-} \right] \left[F \right] \left[e^x \right]$$

$$4.97871$$

$$-2$$

$$10^{2.32} = 208.93$$

$$2 \left[. \right] 32 \left[F \right] \left[10^x \right]$$

$$208.93$$

FACTORIAL

The factorial $n!$ is calculated to ± 2 in the second digit of the mantissa. Values must be positive and in the range 1 to 69. The maximum calculation time for $n = 69$ is 10 s.

WORKED EXAMPLE

$$C_{10}^6 = 10! / (6! \cdot (10 - 6)!) = 210$$

$$10 \left[F \right] \left[n! \right] \div \left[\left[\left[6 \right] \left[F \right] \left[n! \right] \right] \right] \times \left[\left[\left[10 \right] \left[- \right] 6 \right] \right] \left[= \right] \left[F \right] 210.$$

$$\left[n! \right] \left[\right] \left[\right] \left[\right] \left[= \right]$$

Key sequence reproduced exactly as printed in the original.

CONVERTING DEGREES TO RADIANS AND RADIANS TO DEGREES

Conversion between degrees and radians is limited only for arguments whose result would fall outside the calculator's number range. For radians to degrees the

permitted range is $|x| < (\pi/180) \cdot 10^{100}$; for degrees to radians, $|x| > (180/\pi) \cdot 10^{-99}$.

WORKED EXAMPLES

$$\pi \text{ rad} = 180^\circ$$

$$\pi \text{ [F] [P} \rightarrow \Gamma]$$

179.99999

$$180^\circ = \pi \text{ rad}$$

$$180 \text{ [F] [\Gamma} \rightarrow \text{P}]$$

3.1415926

3. SAFETY INSTRUCTIONS

WARNING! Take care when running the microcalculator from the mains through the power supply unit! If the power supply circuit breaks down, the MC may carry 220 V. Always disconnect the power supply from the mains first, then the MC from the power supply.

Do not open the microcalculator or carry out repair or adjustment work while the power supply is connected.

4. BRIEF DESCRIPTION OF THE MICROCALCULATOR

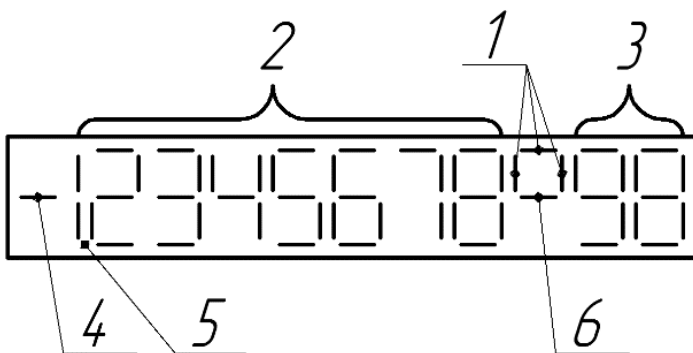
The microcalculator is designed for complex mathematical and engineering calculations by students, engineers and scientists.

It automatically performs the four arithmetic operations; natural and common logarithms and antilogarithms; trigonometric and inverse trigonometric functions; reciprocals; factorials; calculations with two levels of brackets; powers; roots; memory operations; and conversion of degrees to radians and radians to degrees.

Data and commands are entered from the keyboard by pressing keys.

The dual-function key [F] lets keys perform two operations. The second functions are marked on the panel above the keys.

Input data and results are shown on a twelve-digit vacuum fluorescent display, in exponential or fixed-point form. The example below of a number in exponential form reads: minus $1.2345678 \cdot 10^{-98}$.



- 1 memory-in-use indicator
- 2 mantissa
- 3 exponent
- 4 number sign indicator
- 5 decimal point
- 6 exponent sign indicator

KEY FUNCTIONS

Clear key $\boxed{C}$

– If the microcalculator is in dual-function mode, press $\boxed{C}$ three times to return to normal mode and clear the registers. The first press cancels dual-function mode, the second clears the display register, and the third clears the constant register.

– If the microcalculator is in overflow mode, press $\boxed{C}$ once to return to normal mode and clear the registers.

Pressing $\boxed{C}$ does not clear the memory register.

IN NORMAL MODE

$\boxed{0} \dots \boxed{9}$

digit keys;

$\boxed{.}$

decimal point key;

$\boxed{+} \quad \boxed{-} \quad \boxed{\times} \quad \boxed{\div}$

arithmetic keys. Pressing any of these carries out the previously set operation (if any), then prepares the calculator for the operation just entered;

$\boxed{=}$

execute key;

$\boxed{[(]}$	two-level bracket keys;
$\boxed{[-/]}$	change sign of the number or of its exponent;
$\boxed{\leftrightarrow}$	swap the contents of the constant register and the display register;
$\boxed{\pi}$	enter π (the display shows 3.1415926);
$\boxed{B\pi}$	enter exponent (EE).

IN DUAL-FUNCTION MODE

Press $\boxed{F}$ to put the calculator into dual-function mode. The $\boxed{\text{arc}}$ key gives the inverse trigonometric functions when followed by $\boxed{\sin}$, $\boxed{\cos}$ or $\boxed{\text{tg}}$.

$\boxed{y^x}$	raise a number to a power;
$\boxed{\sqrt{}}$	square root;
$\boxed{1/X}$	reciprocal;
$\boxed{n!}$	factorial;
$\boxed{\sin}$ $\boxed{\cos}$ $\boxed{\text{tg}}$	trigonometric functions;
$\boxed{\lg}$ $\boxed{\ln}$ $\boxed{10^x}$ $\boxed{e^x}$	common and natural logarithms and antilogarithms;
$\boxed{\Pi+}$ $\boxed{\Pi-}$ $\boxed{\Pi\times}$ $\boxed{\Pi\div}$	memory arithmetic ($M+$, $M-$, $M\times$, $M\div$);
$\boxed{\Gamma\rightarrow P}$ $\boxed{P\rightarrow\Gamma}$	convert degrees to radians, and radians to degrees;
$\boxed{X\leftrightarrow\Pi}$	swap the display register and the memory register ($X\leftrightarrow M$);
$\boxed{\text{И}\Pi}$	recall memory to the display (MR);
$\boxed{3\Pi}$	store the display register in memory (MS);
$\boxed{C\Pi}$	clear memory (MC);
$\boxed{CF}$	cancel dual-function mode.

Note. The memory operations $\boxed{\Pi+}$, $\boxed{\Pi-}$, $\boxed{\Pi\times}$ and $\boxed{\Pi\div}$ do not change the number on the display; the result is kept in the memory register. Clearing memory does not affect the display register or a pending operation.

IN OVERFLOW MODE

Overflow mode occurs if:

– any result or intermediate result is greater than $9.9999999 \cdot 10^{99}$ or less than 10^{-99} , regardless of sign;

– the result of a memory operation is greater than $9.9999999 \cdot 10^{99}$ or less than 10^{-99} , regardless of sign (the previous memory contents are replaced by the number that caused the overflow);

– a division by zero is attempted;

– a function is evaluated for an argument outside its domain.

The display then shows 10 significant digits, or 10 zeros, with the decimal point lit in every position.

Press  once to clear overflow mode.

Note. The overflow indication may disappear after other keys are pressed, but the calculator will not have returned to normal mode, and later calculations may be wrong.

5. PREPARATION FOR USE AND OPERATING PROCEDURE

5.1. Battery operation.

Remove the cover, insert the batteries into the compartment as marked, and close the battery compartment. Slide the power switch to the right (towards the dot). The display shows “0” in the 8th digit of the mantissa. The microcalculator is ready.

Note. If the display does not light, rotate the batteries a few times about their axis. A marked drop in display brightness means the supply voltage is low and the batteries need replacing.

When finished, slide the power switch to the left (away from the dot).

5.2. Operation from the power supply unit.

Insert the power supply’s plug into the MC’s socket, then plug the power supply into a 220 V mains outlet. Switch on the microcalculator. The display shows “0” in the 8th digit of the mantissa. The microcalculator is ready.

When finished, slide the power switch to the left (away from the dot), unplug the power supply from the mains, then disconnect it from the microcalculator.

Note. The microcalculator has an interlock that disconnects the batteries when it runs from the mains via the power supply.

WARNING! Wait at least 30 s before switching the microcalculator on again. Replace the batteries only with the power supply disconnected. Do not use batteries past their expiry date.

Check that the microcalculator works correctly by entering the control test in the table.

Table

Step	Key sequence	Display			
		Number sign	Mantissa 8 7 6 5 4 3 2 1 (digit)	Exponent	
				Sign	2 1
1.	1 2 3 4 5 6 7 9				
2.	$\times$ 81 $\div$				
	=	–	9.9 9 9 9 9 9 9		8
3.	$\div$ 99999999		9 9 9 9 9 9 9 9.		
4.	+ 40 =		30.		
$\overline{P/T}$ switch set to $\overline{\Gamma}$					
5.	F sin		5.	–	1
6.	arc cos		6 0.		
7.	– 15 =		4 5.		
8.	F tg		1.		
9.	F $P \rightarrow \Gamma$		5 7. 2 9 5 7 7 9		
10.	F $\Gamma \rightarrow P$		1.		
11.	BΠ		1.		0 0
12.	F 3Π		1.		
13.	F CΠ		1.		
14.	$\leftrightarrow$		1 5.		
15.	π		3. 1 4 1 5 9 2 6		
16.	$\leftrightarrow$		1.		
“Elektronika MK 66”: $\overline{P/T}$ switch set to $\overline{P}$					
17.	arc tg		7. 8 5 3 9 8	–	1

Step	Key sequence	Display			
		Number sign	Mantissa 8 7 6 5 4 3 2 1 (digit)	Exponent	
				Sign	2 1
18.	$\boxed{F} \boxed{tg}$		1.		
19.	1 1 1 1 1 1 1 1		1 1 1 1 1 1 1 1.		
20.	$\boxed{F} \boxed{n!}$	.	4. 3. 7. 1. 2. 0. 4. 5.		9. 0.
<i>“Elektronika MK 66A”</i>					
17.	$\boxed{arc} \boxed{tg} \boxed{F}$				
	$\boxed{\Gamma \rightarrow P}$		7. 8 5 3 9 8 1 7	–	1
18.	$\boxed{F} \boxed{P \rightarrow \Gamma}$				
	$\boxed{F} \boxed{tg}$		1.		
19.	1 1 1 1 1 1 1 1		1 1 1 1 1 1 1 1.		
20.	$\boxed{F} \boxed{n!}$	.	4. 3. 7. 1. 2. 0. 4. 5.		9. 0.

Note. With a modified display, the dot in the “number sign” position may be absent.

6. TRANSPORT AND STORAGE

When transported, the packed microcalculator must be protected from precipitation and mechanical damage.

After transport in winter conditions, keep the microcalculator in its packaging at room temperature for at least 1 hour.

Store the microcalculator at an air temperature of 1 to 40 °C, a relative humidity of no more than 90 %, and with no acidic, alkaline or other aggressive impurities in the air.

STORING BATTERIES INSIDE THE MICROCALCULATOR MAY CAUSE ELECTROLYTE LEAKAGE AND DAMAGE THE MC.

7. WARRANTY

The microcalculator conforms to the approved design and meets the requirements of technical specifications 6K0.310.066 TU.

The warranty period is 24 months from the date of sale. If the warranty card and tear-off coupons have no mark from the retailer, the period is counted from the date the microcalculator left the factory.

If your calculator needs repair, please write its serial number (marked on the base) and the MC type on the warranty card and tear-off coupon of this manual.

Quality claims will not be accepted, and warranty repair will not be carried out:

- if the warranty card is not presented;
- if the seals on the MC or the power supply are broken;
- if the operating conditions have been breached (mechanical damage to the MC, batteries stored inside the MC, or batteries of another type used).

If the microcalculator malfunctions, the owner is entitled to free repair during the warranty period. After repair, the tear-off coupon stays with the workshop.

If manufacturing defects are found, the owner is entitled to a replacement within the warranty period.

Replacement is made through the retail network on the report of the repair organisation, in accordance with the current exchange rules.

ATTENTION! The manufacturer is not responsible for the quality of the batteries. Send all battery quality claims directly to their manufacturers.

8. WARRANTY WORKSHOP INFORMATION

Manufacturer's workshop:

432030, Ulyanovsk, "Iskra" Plant, 2 Repina St. Telephone 34-93-04.

Note. Addresses of warranty and repair workshops are available from any shop that sells microcalculators.

When sending the microcalculator to the manufacturer's workshop, package it so it cannot be mechanically damaged in transit. Include the sender's full postal address.

WARRANTY CARD AND TEAR-OFF COUPONS

The original manual ends with a warranty card and two tear-off repair coupons. Their wording is translated below; the blank lines were for handwritten entries and stamps.

Valid when completed

Free wholesale price with power supply: 72 rubles

without power supply: 65 rubles

Protocol No. 17/91 of 14.06.91

Retail markup

Free retail price

WARRANTY CARD

To be completed by the manufacturer

Microcalculator "Elektronika MK 66" No. _____

Date of manufacture _____

Quality control (OTK) representative _____ *OTK stamp*

Address for quality claims:

432030, Ulyanovsk, "Iskra" Plant, 2 Repina St.

To be completed by the retailer

Date of sale _____ (day, month, year)

Seller _____ (signature or stamp)

Shop stamp

Valid when completed

**TEAR-OFF COUPON FOR WARRANTY REPAIR
IN THE FIRST YEAR OF WARRANTY**

(A second, identical coupon covers the second year. Each coupon has a stub along the cut line, labelled "Stub of tear-off coupon for warranty repair in the first [second] year of warranty".)

To be completed by the manufacturer

Microcalculator "Elektronika MK 66 " No. _____

Date of manufacture _____

Quality control (OTK) representative _____ *OTK stamp*

Address for returning the coupon to the manufacturer:

432030, Ulyanovsk, "Iskra" Plant, 2 Repina St.

To be completed by the retailer

Date of sale _____ (day, month, year)

Seller _____ (signature or stamp)

Shop stamp

To be completed by the repair organisation

Valid when completed

Microcalculator warranty number _____

Repair work carried out _____

Date of repair _____ (day, month, year)

Signature of person who carried out the repair _____

Signature of calculator owner confirming the repair _____

Repair organisation stamp, with town

REPAIR ORGANISATION SECTION (stub side)

To be completed by the repair organisation

Registered for warranty service _____ (name of repair organisation)

_____ (day, month, year)

Warranty repair: _____

TRANSLATOR'S NOTE: PANEL LABELS

Key and switch labels are shown as they appear on the calculator, in Cyrillic, so they match your keyboard. Their meanings:

Γ/P P/Γ

Degrees / Radians switch ($\Gamma = \textit{gradusy}$, degrees; $\text{P} = \textit{radiany}$, radians)

$\Gamma \rightarrow \text{P}$ $\text{P} \rightarrow \Gamma$

degrees $\rightarrow$ radians; radians $\rightarrow$ degrees

$\text{B}\Pi$

enter exponent (EE / EXP)

$\Pi+$ $\Pi-$ $\Pi\times$ $\Pi\div$

memory arithmetic, $\text{M}+$ $\text{M}-$ $\text{M}\times$ $\text{M}\div$ ($\Pi = \textit{pamyat'}$, memory)

3Π

memory store (MS)

$\text{И}\Pi$

memory recall (MR)

$\text{C}\Pi$

memory clear (MC)

$\text{X} \leftrightarrow \Pi$

exchange display and memory ($\text{X} \leftrightarrow \text{M}$)

tg

tangent (tan)

lg

common (base-10) logarithm (log)

In the worked examples, the right-hand columns show the display: mantissa, then exponent. For example, “5. -1 ” means $5 \times 10^{-1} = 0.5$. Decimal commas in the original are shown as decimal points.